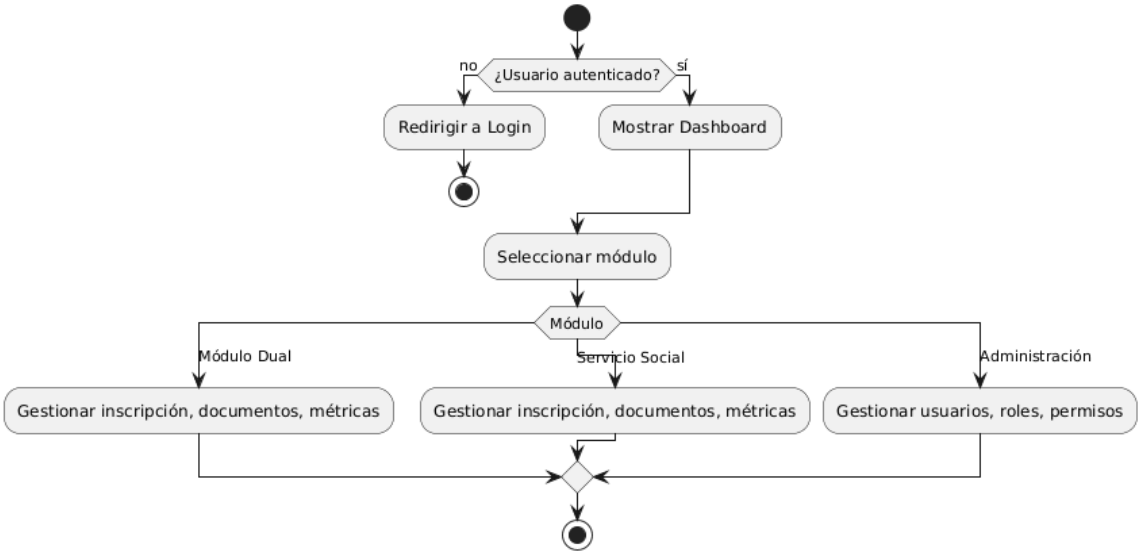
1. User Flow (UX) – PlantUML

```
@startuml
start
:Inicio;
if (¿Autenticado?) then (no)
  :Login;
endif
:Dashboard;
partition "Módulo Dual" {
  :Ver programas duales;
  :Registrar inscripción a módulo dual;
  :Subir documentos;
  :Consultar avance y métricas;
}
partition "Servicio Social" {
  :Ver programas de servicio social;
  :Registrar inscripción a servicio social;
  :Subir documentos;
  :Consultar avance y métricas;
}
:Gestión de usuarios, roles y permisos;
:Logout;
stop
@enduml
```

Inicio
¿Autenticado?
no
Login
Dashboard
Módulo Dual
Ver programas duales
Registrar inscripción a módulo dual
Subir documentos
Consultar avance y métricas
Servicio Social
Ver programas de servicio social
Registrar inscripción a servicio social
Subir documentos
Consultar avance y métricas
Gestión de usuarios, roles y permisos
Logout

2. Flowchart (General) – PlantUML

```
@startuml
start
if (¿Usuario autenticado?) then (no)
  :Redirigir a Login;
  stop
else (sí)
  :Mostrar Dashboard;
endif
:Seleccionar módulo;
switch (Módulo)
  case (Módulo Dual)
    :Gestionar inscripción, documentos, métricas;
  case (Servicio Social)
    :Gestionar inscripción, documentos, métricas;
  case (Administración)
    :Gestionar usuarios, roles, permisos;
endswitch
stop
@enduml
```

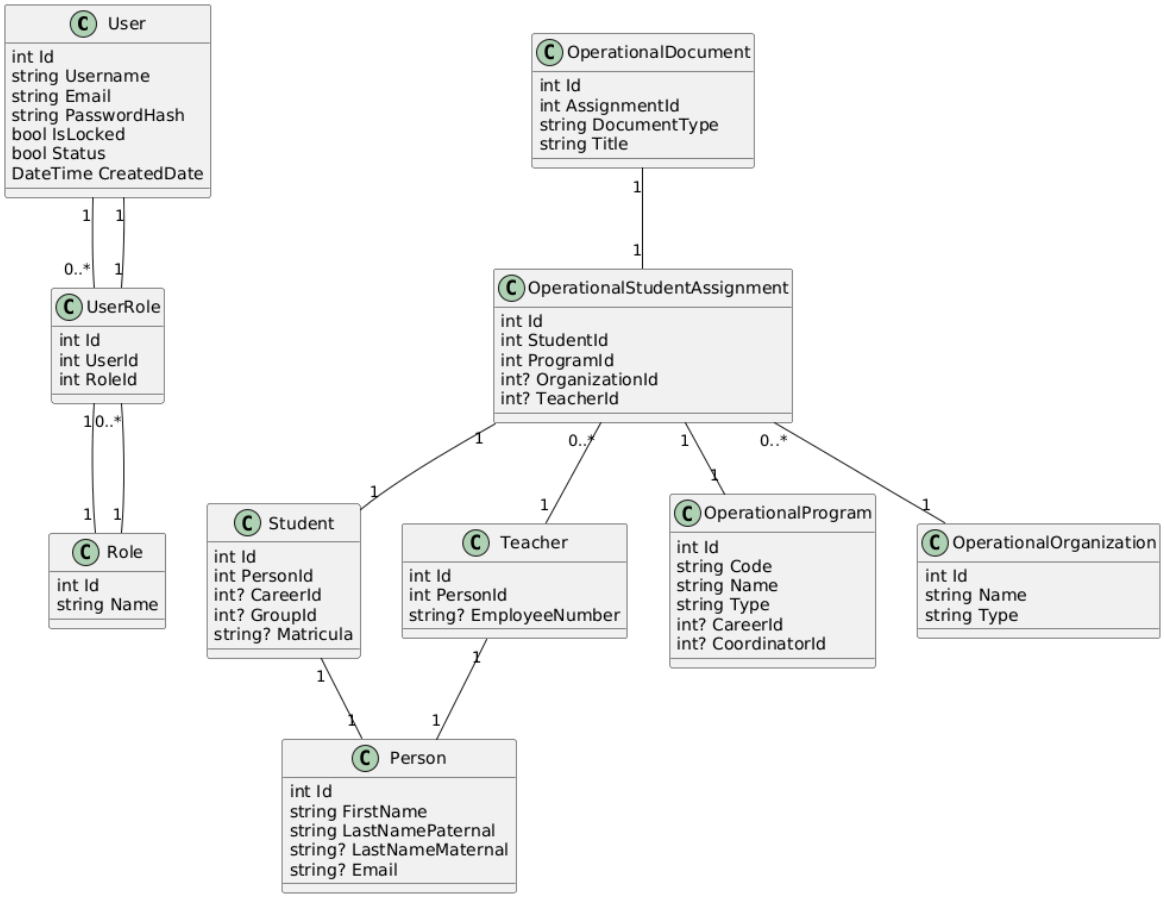

¿Usuario autenticado?
no
sí
Redirigir a Login
Mostrar Dashboard
Seleccionar módulo
Módulo
Módulo Dual
Servicio Social
Administración
Gestionar inscripción, documentos, métricas
Gestionar inscripción, documentos, métricas
Gestionar usuarios, roles, permisos

3. Structure Diagram (UML) – PlantUML (Clases)

```
@startuml
class User {
  int Id
  string Username
  string Email
  string PasswordHash
  bool IsLocked
  bool Status
  DateTime CreatedDate
}
class Role {
  int Id
  string Name
}
class UserRole {
  int Id
  int UserId
  int RoleId
}
class Student {
  int Id
  int PersonId
  int? CareerId
  int? GroupId
  string? Matricula
}
class Teacher {
```

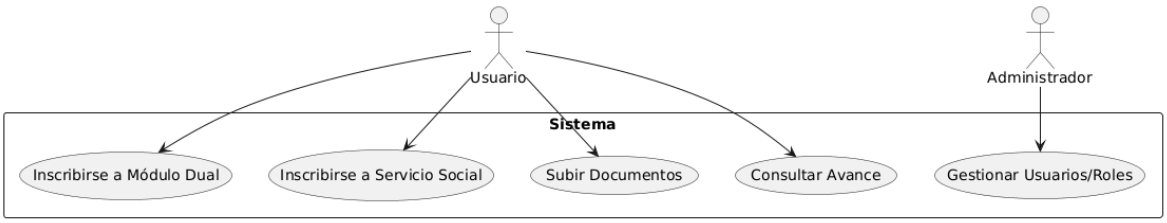
```
  int Id
  int PersonId
  string? EmployeeNumber
}
class Person {
  int Id
  string FirstName
  string LastNamePaternal
  string? LastNameMaternal
  string? Email
}
class OperationalProgram {
  int Id
  string Code
  string Name
  string Type
  int? CareerId
  int? CoordinatorId
}
class OperationalStudentAssignment {
  int Id
  int StudentId
  int ProgramId
  int? OrganizationId
  int? TeacherId
}
class OperationalOrganization {
  int Id
  string Name
```

```
  string Type
}
class OperationalDocument {
  int Id
  int AssignmentId
  string DocumentType
  string Title
}

User "1" -- "0..*" UserRole
Role "1" -- "0..*" UserRole
UserRole "1" -- "1" User
UserRole "1" -- "1" Role
Student "1" -- "1" Person
Teacher "1" -- "1" Person
OperationalStudentAssignment "1" -- "1" Student
OperationalStudentAssignment "1" -- "1" OperationalProgram
OperationalStudentAssignment "0..*" -- "1" OperationalOrganization
OperationalStudentAssignment "0..*" -- "1" Teacher
OperationalDocument "1" -- "1" OperationalStudentAssignment
@enduml
```


User
int Id
string Username
string Email
string PasswordHash
bool IsLocked
bool Status
DateTime CreatedDate
UserRole
int Id
int UserId
int RoleId
Role
int Id
string Name
OperationalDocument
int Id
int AssignmentId
string DocumentType
string Title
OperationalStudentAssignment
int Id
int StudentId
int ProgramId
int? OrganizationId
int? TeacherId
Student
int Id
int PersonId
int? CareerId
int? GroupId
string? Matricula
Teacher
int Id
int PersonId
string? EmployeeNumber
OperationalProgram
int Id
string Code
string Name
string Type
int? CareerId
int? CoordinatorId
OperationalOrganization
int Id
string Name
string Type
Person
int Id
string FirstName
string LastNamePaternal
string? LastNameMaternal
string? Email
1
0..*

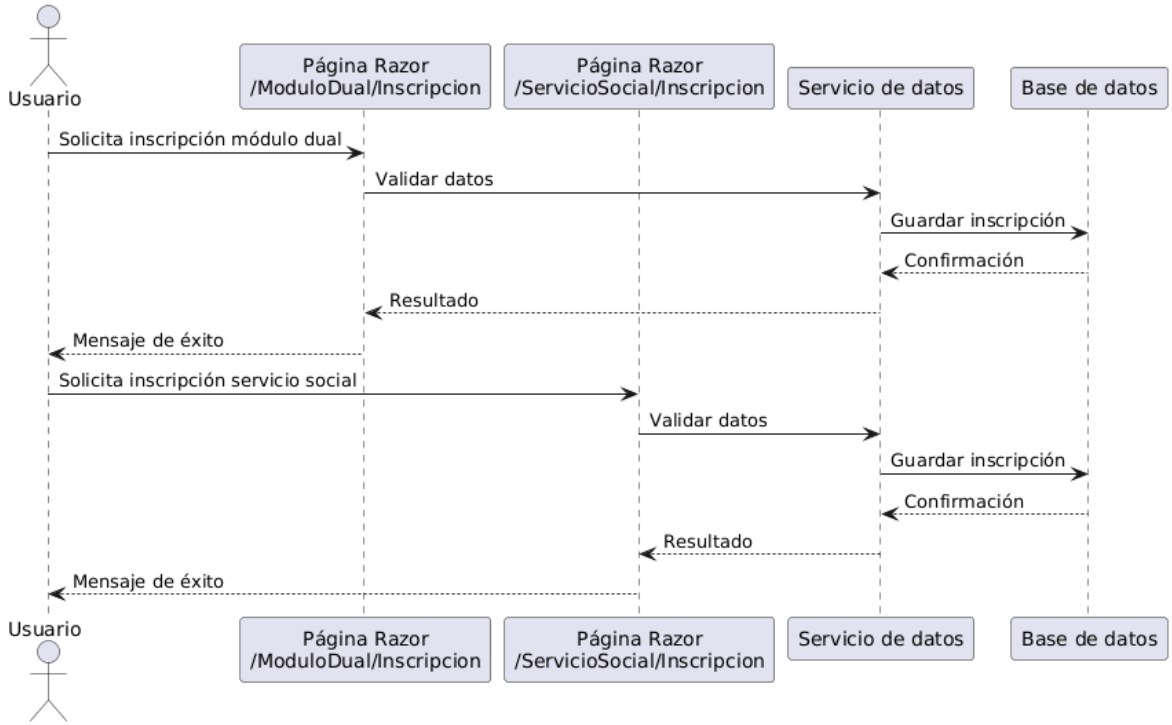
4. Behavior Diagram (UML) – PlantUML (Casos de uso)

```
@startuml
actor Usuario
actor Administrador
rectangle "Sistema" {
  usecase "Inscribirse a Módulo Dual" as UC1
  usecase "Inscribirse a Servicio Social" as UC2
  usecase "Subir Documentos" as UC3
  usecase "Consultar Avance" as UC4
  usecase "Gestionar Usuarios/Roles" as UC5
}
Usuario --> UC1
Usuario --> UC2
Usuario --> UC3
Usuario --> UC4
Administrador --> UC5
@enduml
```



5. Interaction Diagram (UML) – PlantUML (Secuencia)

```
@startuml
actor Usuario
participant "Página Razor\n/ModuloDual/Inscripcion" as PaginaDual
participant "Página Razor\n/ServicioSocial/Inscripcion" as PaginaSS
participant "Servicio de datos" as Servicio
participant "Base de datos" as DB

Usuario -> PaginaDual : Solicita inscripción módulo dual
PaginaDual -> Servicio : Validar datos
Servicio -> DB : Guardar inscripción
DB --> Servicio : Confirmación
Servicio --> PaginaDual : Resultado
PaginaDual --> Usuario : Mensaje de éxito

Usuario -> PaginaSS : Solicita inscripción servicio social
PaginaSS -> Servicio : Validar datos
Servicio -> DB : Guardar inscripción
DB --> Servicio : Confirmación
Servicio --> PaginaSS : Resultado
PaginaSS --> Usuario : Mensaje de éxito
@enduml
```

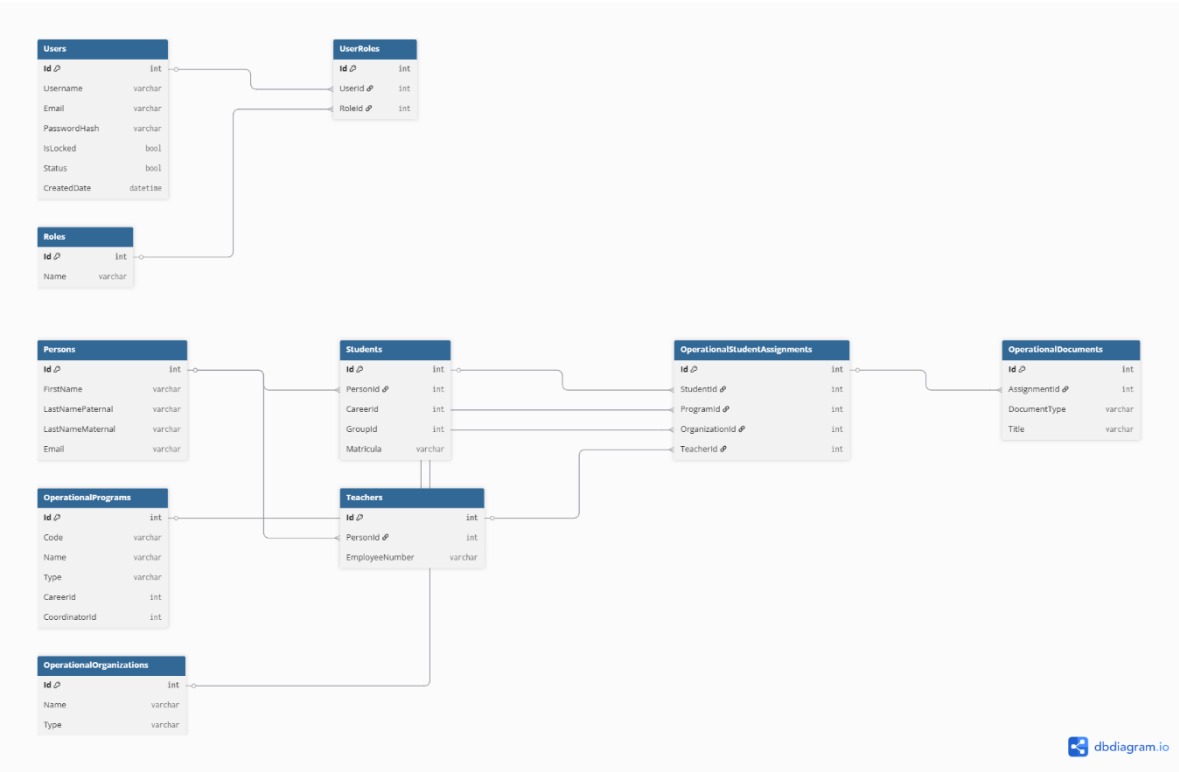

Usuario
Página Razor /ModuloDual/Inscripcion
Página Razor /ServicioSocial/Inscripcion
Servicio de datos
Base de datos
Solicita inscripción módulo dual
Validar datos
Guardar inscripción
Confirmación
Resultado
Mensaje de éxito
Solicita inscripción servicio social
Validar datos
Guardar inscripción
Confirmación
Resultado
Mensaje de éxito
Usuario
Página Razor /ModuloDual/Inscripcion
Página Razor /ServicioSocial/Inscripcion
Servicio de datos
Base de datos

6. Relational Model (DBMS) – dbdiagram.io

```
Table Users {
  Id int [pk]
  Username varchar
  Email varchar
  PasswordHash varchar
  IsLocked bool
  Status bool
  CreatedDate datetime
}

Table Roles {
  Id int [pk]
  Name varchar
}

Table UserRoles {
  Id int [pk]
  UserId int
  RoleId int
}

Table Persons {
  Id int [pk]
  FirstName varchar
  LastNamePaternal varchar
  LastNameMaternal varchar
```

```
  Email varchar
}

Table Students {
  Id int [pk]
  PersonId int
  CareerId int
  GroupId int
  Matricula varchar
}

Table Teachers {
  Id int [pk]
  PersonId int
  EmployeeNumber varchar
}

Table OperationalPrograms {
  Id int [pk]
  Code varchar
  Name varchar
  Type varchar
  CareerId int
  CoordinatorId int
}

Table OperationalStudentAssignments {
  Id int [pk]
  StudentId int
```

```
  ProgramId int
  OrganizationId int
  TeacherId int
}

Table OperationalOrganizations {
  Id int [pk]
  Name varchar
  Type varchar
}

Table OperationalDocuments {
  Id int [pk]
  AssignmentId int
  DocumentType varchar
  Title varchar
}

Ref: UserRoles.UserId > Users.Id
Ref: UserRoles.RoleId > Roles.Id
Ref: Students.PersonId > Persons.Id
Ref: Teachers.PersonId > Persons.Id
Ref: OperationalStudentAssignments.StudentId > Students.Id
Ref: OperationalStudentAssignments.ProgramId > OperationalPrograms.Id
Ref: OperationalStudentAssignments.OrganizationId > OperationalOrganizations.Id
Ref: OperationalStudentAssignments.TeacherId > Teachers.Id
Ref: OperationalDocuments.AssignmentId > OperationalStudentAssignments.Id
```


Users
Id int
Username varchar
Email varchar
PasswordHash varchar
IsLocked bool
Status bool
CreatedDate datetime
UserRoles
Id int
UserId int
RoleId int
Roles
Id int
Name varchar
Persons
Id int
FirstName varchar
LastNamePaternal varchar
LastNameMaternal varchar
Email varchar
Students
Id int
PersonId int
CareerId int
GroupId int
Matricula varchar
OperationalStudentAssignments
Id int
StudentId int
ProgramId int
OrganizationId int
TeacherId int
OperationalDocuments
Id int
AssignmentId int
DocumentType varchar
Title varchar
OperationalPrograms
Id int
Code varchar
Name varchar
Type varchar
CareerId int
CoordinatorId int
Teachers
Id int
PersonId int
EmployeeNumber varchar
OperationalOrganizations
Id int
Name varchar
Type varchar
dbdiagram.io